

Mathematics

A Cultural Heritage

Lecture 2

**Numbers:
Live Discussion**

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

*Mathematics
and the Arts*

Mathematics
and Education

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

Logistics: Continued

1. Any concerns about assignments?
2. Please contact us if there are issues (tutorial registration, technical issues, etc.)
3. Course topics
4. General questions about the course logistics?

Tutorial registration

1. As of 9am today, there were 6 students not registered.
2. Contact us if you are having trouble registering.
3. If tutorials don't fit your schedule, consider registering for another course (you are welcome next year!).

Numbers on a fundamental level

Numbers



Bank Account Information		
Starting Balance	01/12/2014	23000
Deposit	07/12/2014	1100
Withdrawal	13/12/2014	1300
New Balance	01/01/2015	22800

What about “-1300”?

What does “3” mean?

Discussion: Number systems

Number Systems of different cultures

	1	2	3	4	5	6	7	8	9
UNITS	𐍌	𐍌𐍌	𐍌𐍌𐍌	𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌
TENS	𐍌𐍌	𐍌𐍌𐍌	𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌
HUNDREDS	𐍌𐍌𐍌	𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌
THOUSANDS	𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌	𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌
TENS OF THOUSANDS	𐍌𐍌𐍌𐍌𐍌								
HUNDREDS OF THOUSANDS	𐍌𐍌𐍌𐍌𐍌𐍌								

Egyptian hieratic numerals of the New Kingdom (1600–1000bc) used more symbols than before, making numbers more compact but harder to learn to use.

1	I	21	XXI	41	XLI	61	LXI	81	LXXXI
2	II	22	XXII	42	XLII	62	LXII	82	LXXXII
3	III	23	XXIII	43	XLIII	63	LXIII	83	LXXXIII
4	IV	24	XXIV	44	XLIV	64	LXIV	84	LXXXIV
5	V	25	XXV	45	XLV	65	LXV	85	LXXXV
6	VI	26	XXVI	46	XLVI	66	LXVI	86	LXXXVI
7	VII	27	XXVII	47	XLVII	67	LXVII	87	LXXXVII
8	VIII	28	XXVIII	48	XLVIII	68	LXVIII	88	LXXXVIII
9	IX	29	XXIX	49	XLIX	69	LXIX	89	LXXXIX
10	X	30	XXX	50	L	70	LXX	90	XC
11	XI	31	XXXI	51	LI	71	LXXI	91	XCI
12	XII	32	XXXII	52	LII	72	LXXII	92	XCH
13	XIII	33	XXXIII	53	LIII	73	LXXIII	93	XCHH
14	XIV	34	XXXIV	54	LIV	74	LXXIV	94	XCIV
15	XV	35	XXXV	55	LV	75	LXXV	95	XCv
16	XVI	36	XXXVI	56	LVI	76	LXXVI	96	XCVI
17	XVII	37	XXXVII	57	LVII	77	LXXVII	97	XCvII
18	XVIII	38	XXXVIII	58	LVIII	78	LXXVIII	98	XCvIII
19	XIX	39	XXXIX	59	LIX	79	LXXIX	99	XCIX
20	XX	40	XL	60	LX	80	LXXX	100	C

200 CC 300 CCC 400 CD 500 D 600 DC 700 DCC 800 DCCC 900 CM 1000 M

0	〇	零			
1	一	壹	10	十	拾
2	二	貳	20	廿/廿	貳拾
3	三	叁	30	卅	參拾
4	四	肆	40	卌	肆拾
5	五	伍			
6	六	陸	10 ²	百	佰
7	七	柒	10 ³	千	仟
8	八	捌	10 ⁴	万	萬
9	九	玖	10 ⁸	亿	億

©G. Szecsenyi, www.archimedes-lab.org

Possible Progression										
0	1	2	3	4	5	6	7	8	9	
0	1	2	3	4	5	6	7	8	9	
0	1	2	3	4	5	6	7	8	9	

www.MrGeri.com

Numbers and generalization

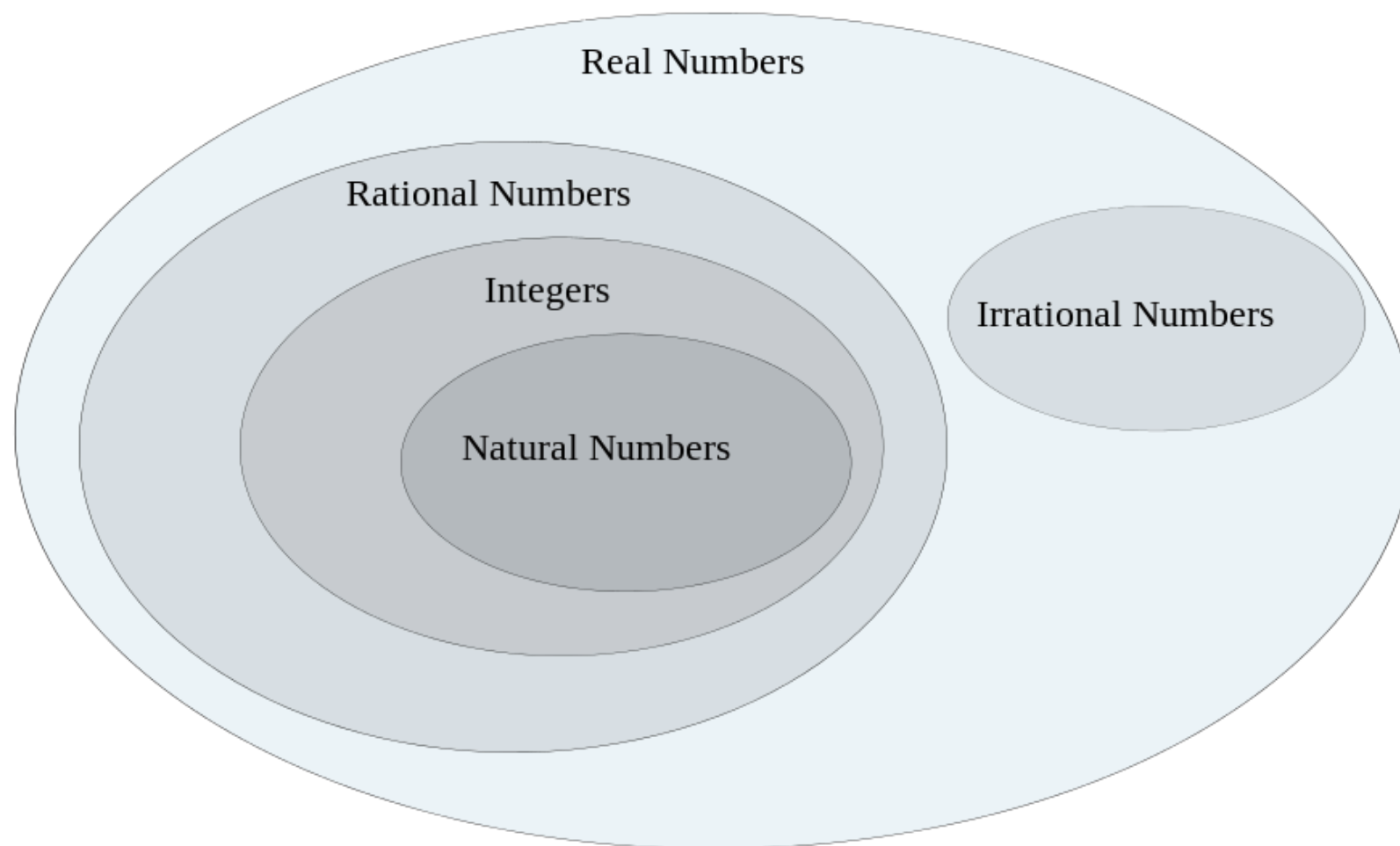
Different kinds of numbers

- Whole/natural numbers: **1, 2, 3, ...**
- Fractions/rational numbers: $\frac{1}{2}, \frac{4}{9}, \frac{11}{13}, \dots$
- Irrational numbers: $\sqrt{7}, \pi \approx 3.14159 \dots$
- Negative numbers: **-3, -7, -10, ...**
- The number **0**
- Imaginary numbers: $i = \sqrt{-1}$

Numbers and generalization

- Complex Numbers: $3 + 2\sqrt{-1}, 7 - i$
- Matrices: $\begin{bmatrix} 3 & 7 \\ 4 & 2 \end{bmatrix}, \begin{bmatrix} \frac{5}{3} & \frac{2}{7} \\ 4 & 0 \end{bmatrix}$
- Number fields, Quaternions: ???
- Infinity: ∞

Types of Numbers



"Real numbers" by Damien Karras - Own work.
Licensed under Public Domain via Wikimedia Commons

Questions/Discussion

1. How do we process numbers?
2. Do numbers have cultural meaning or significance?
3. What do we do with numbers? Why are they important?

This Week's Survey

1. On a scale of 0 (extremely low) to 10 (extremely high), rate your own mathematical ability by circling your choice:

[extremely low] **0 1 2 3 4 5 6 7 8 9 10** [extremely high]

2. Write a few words on the reason of why you achieved such a level of mathematical ability (e.g., I find mathematics boring, or I had some inspiring teachers, etc.)

<https://gear.cetl.hku.hk/survey/9091854a3dae4a379ff81db14c3058b3>

